

The conjectured recurrence of OEIS A034267, from the closed form the entry states

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Abstract

OEIS A034267 states a closed form as a fact and conjectures a P-recursive recurrence. The closed form is hypergeometric – the ratio of consecutive terms is a rational function of n – so the sequence satisfies an exact first-order recurrence with polynomial coefficients, and the conjectured recurrence is decided by reducing it with that ratio. What comes out is an identity in $\mathbb{Q}(n)$, not an agreement of finitely many terms, and no generating function is needed: this entry states none.

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1 What the entry says

The first terms are 0, 1, 7, 39, 196, 930, 4257, 19019, ..., with offset 0. The entry states, as a fact and not as a conjecture,

$$a(n) = \text{binomial}(2n, n+1) \cdot (n^2 + n + 1) / (n+2);$$

that is, $a(n) = \frac{(n^2+n+1)\binom{2n}{n+1}}{n+2}$, and separately records

Conjecture D-finite with recurrence $-(n+2) \cdot (11n-7) \cdot a(n) + 2 \cdot (23n^2+44n+30) \cdot a(n-1) - 4 \cdot (n+5) \cdot (2n-3) \cdot a(n-2) = 0$. - R. J. Mathar, Feb 10 2025

2 The closed form is hypergeometric

Directly from the expression,

$$\frac{a(n)}{a(n-1)} = \frac{2n(2n-1)(n^2+n+1)}{(n-1)(n+2)(n^2-n+1)},$$

a rational function of n . Equivalently a satisfies the exact first-order recurrence

$$(n^4 - 2n^2 + 3n - 2) a(n) = (4n^4 + 2n^3 + 2n^2 - 2n) a(n-1),$$

whose leading coefficient is a nonzero polynomial, so it determines every term past the finitely many zeros of that polynomial from the one before it.

3 The conjectured recurrence follows

Substituting $a(n) = r(n)a(n-1)$ and $a(n-1) = r(n-1)a(n-2)$ into the conjectured relation and dividing through by $a(n-2)$ leaves a rational function of n alone. That function **simplifies to zero identically**, so the conjectured relation holds wherever the terms it names are defined – at every index, not merely at those checked.

Theorem 1. *The conjecture recorded in OEIS A034267 is true.*

Proof. The displayed reduction is an identity in $\mathbb{Q}(n)$ between the left-hand side of the conjecture divided by $a(n-2)$ and zero. Since $a(n-2) \neq 0$ for the indices concerned, the conjecture itself holds there. $\square$

4 What this rests on

On the entry's own closed form and nothing else. That statement is not taken on trust: the expression was evaluated and agrees with all 26 terms the entry publishes. It was read from the entry's factual formulas only – a closed form inside a conjectural block, or one qualified by a finite range, is itself a conjecture and is not a premise.